

DEPARTMENT OF MATHEMATICS, GTIIT.
MIDTERM - CALCULUS 1 M

Mark your answer for each question with ✓ at the appropriate item in the question, every question has exactly one correct answer. You don't need to justify or prove your answers unless you are asked to. In this exam there are 10 questions. All answers should be on these questions sheets.

[2%] (1) The function $f(x) = x \cdot D(x)$ where D is Dirichlet's function

1. Has no limit in any point.
2. Has a limit only in $x = 0$.
3. Has a limit in a finite number of points.
4. Has a limit only in $x \in \mathbb{Q}$.
5. None of the above.

Does $f(x) = x \cdot D(x)$ have a limit in $x = 0$? YES / NO. Prove your answer

[1%] (2) The multiplication of any two irrational numbers is

1. a rational number.
2. an irrational number.
3. an integer.
4. not an integer.
5. None of the above.

[1%] (3) A student proved that $(\forall \epsilon)(\exists \delta)(\forall x, 0 < |x - x_0| < \delta) |f(x) - L| \leq \epsilon$. The definition we have in class is $(\forall \epsilon)(\exists \delta)(\forall x, 0 < |x - x_0| < \delta) |f(x) - L| < \epsilon$. The student's proof implies

1. $\lim_{x \rightarrow x_0} f(x) = L$.
2. $\lim_{x \rightarrow x_0} f(x) < L$.
3. $\lim_{x \rightarrow x_0} f(x) > L$.
4. $\lim_{x \rightarrow x_0} f(x) \neq L$.
5. None of the above.

[1%] (4) For $a > 0$ the rule $\frac{a}{0^-} = -\infty$ means

Theorem

[2%] (5) Find

$$\lim_{x \rightarrow -\infty} \frac{x^3 - 3x^2}{(\pi/2 + \arctan x)(4 - x^3)} =$$

1. Not Exist.
2. ∞ .
3. $-\infty$.
4. $-1/\pi$.
5. $1/\pi$.
6. None of the above.

[1%] (6) Find

$$\lim_{x \rightarrow \infty} 2\sqrt{x}(\sqrt{4x-1} - \sqrt{4x+1}) =$$

1. Not Exist.
2. ∞ .
3. $-\infty$.
4. -1 .
5. $+1$.
6. None of the above.

[1%] (7) Write the definition of $\lim_{x \rightarrow -\infty} f(x) = -\infty$

[3%] (8) Find

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n} - 1 \right)^n =$$

1. Not Exist.
2. e .
3. $-e$.
4. e^{-1} .
5. $-e^{-1}$.
6. -1 .
7. $+1$.
8. ∞ .
9. $-\infty$.
10. None of the above.

Proof (for question (8))

[1%] (9) State Bolzano-Weierstrass Theorem (do not prove it)

[3%] (10) Find

$$\lim_{n \rightarrow \infty} \left(\frac{n+2}{n-3} \right)^n.$$

Answer:

Proof